SmartRepair AI

Help Guide: How to Run the App

A step-by-step guide for first-time users. No programming experience needed.

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1. What you need

SmartRepair AI is a **Streamlit** app: a Python program that opens in your web browser. It runs entirely on your own computer; no internet account is needed once it is installed.

Before you start, check that you have:

Item	Details
A computer	Windows 10/11 or macOS. At least 8 GB of RAM is recommended because the dataset has about 300,000 records.
The SmartRepair folder	Unzipped (not opened inside a .zip file) and saved somewhere easy to find, such as Downloads or Desktop.
Internet connection	Only needed once, to download Python and the packages.
A web browser	Chrome, Edge, Firefox or Safari.

The folder must contain these files. Do not rename or move them:

File	What it is
app.py	The app itself (the file you run).
requirements.txt	The list of Python packages the app needs.
cleaned_repairs_en.csv	The repair dataset. **problem_en** holds the English problem description.
cleaned_categories (1).csv	Names for each product category number.
best_repair_model.joblib	The trained prediction model.
model_*.csv, model_training_info.json, member2_model_comparison.csv	Model evaluation results.
member3_clustered_repairs.csv	Fault-cluster results.
smoke_test.py	An optional check that all files are working.
README.md	Technical notes.

> **Note:** The **user_data** folder does not exist at first. The app creates it automatically when you add records or retrain the model.

2. Install Python (one time only)

Python is the programming language the app is written in. You only install it once.

Windows

1. Go to **https://www.python.org/downloads/** and click the yellow **Download Python 3.12** (or newer) button.

2. Open the downloaded installer.

3. **Important:** on the first screen, tick the box **"Add python.exe to PATH"** at the bottom.

4. Click **Install Now** and wait until it says *Setup was successful*.

> **Common mistake:** If you forget to tick "Add python.exe to PATH", the `python` command will not work. Run the installer again, choose **Modify**, and turn on **Add Python to environment variables**.

macOS

1. Go to **https://www.python.org/downloads/** and download the macOS installer.
2. Open the .pkg file and click **Continue** until it is installed.
3. On macOS, always type **`python3`** instead of **`python`** in the commands below.

Check that Python works

Open a terminal (see section 3) and type the command below, then press **Enter**:

Windows

```
python --version
```

macOS

```
python3 --version
```

You should see something like `Python 3.12.4`. Any version from 3.10 to 3.13 works.

3. Open a terminal in the project folder

A **terminal** is a window where you type commands. The commands must run **inside the SmartRepair folder**.

Windows (PowerShell)

1. Open the SmartRepair folder in **File Explorer** (you should see app.py).
2. Click in the **address bar** at the top, delete the text, type **`powershell`** and press **Enter**.
3. A blue or black window opens. Its prompt ends with the folder name, for example `PS C:\Users\you\Downloads\SmartRepair>`

macOS (Terminal)

1. Open **Terminal** (press Cmd + Space, type *Terminal*, press Enter).
2. Type **`cd`** (c, d, then a space). Do not press Enter yet.
3. Drag the SmartRepair folder from Finder into the Terminal window, then press **Enter**.

Check you are in the right place

Windows

```
dir
```

macOS

```
ls
```

You should see **app.py** in the list. If not, you are in the wrong folder; repeat this section.

4. Install the required packages (one time only)

The app needs a few extra Python packages (Streamlit, pandas, scikit-learn and others). They are listed in requirements.txt. In the terminal, type:

Windows

```
python -m pip install -r requirements.txt
```

macOS

```
python3 -m pip install -r requirements.txt
```

This downloads and installs everything. It can take 2 to 5 minutes. Lots of text will scroll past; that is normal. Wait until the prompt appears again. The last lines usually start with *Successfully installed ...*

Optional: check that everything works

Windows (use python3 on macOS)

```
python smoke_test.py
```

Every line should start with **PASS**, ending with *Smoke test complete*. If a line starts with **FAIL**, see Troubleshooting (section 8).

5. Start the app

In the same terminal, type:

Windows

```
python -m streamlit run app.py
```

macOS

```
python3 -m streamlit run app.py
```

What happens next:

1. **First time only:** Streamlit may ask for your email address. You can leave it empty: just press **Enter**.

2. The terminal shows *You can now view your Streamlit app in your browser* and a **Local URL**, normally **http://localhost:8501**.

3. Your browser opens the app automatically. If it does not, copy the Local URL into your browser's address bar.

4. The first load takes 10 to 30 seconds while the dataset is read. You will see *Loading repair dataset...*

> **Important: Keep the terminal window open** while you use the app. Closing it stops the app.

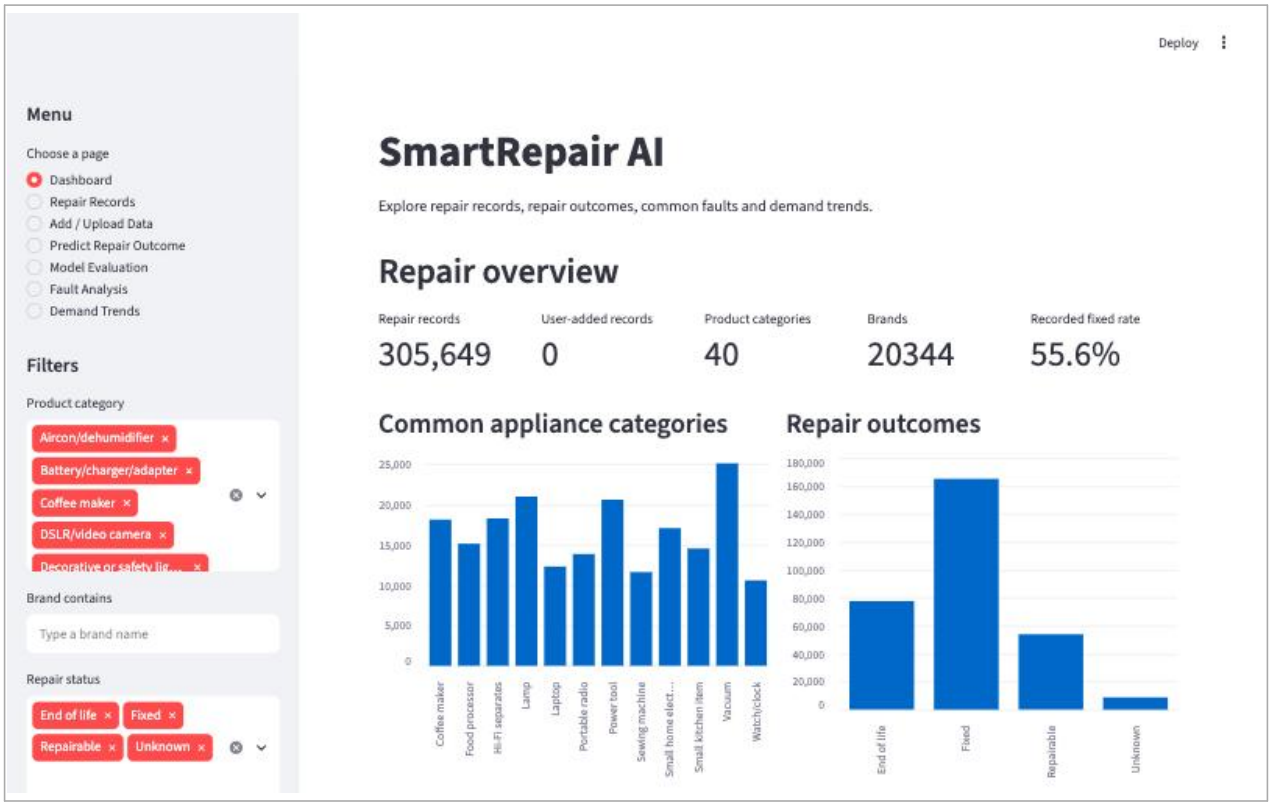


Figure 1: The app after it starts, showing the Dashboard page.

6. Use the app

Use the **Menu** on the left to move between pages. The **Filters** below the menu (product category, brand, repair status and data source) change what every page shows.

Page	What it does
Dashboard	Summary numbers and charts: most common categories, brands, repair outcomes, trends by year and product age.
Repair Records	Browse and search the records (search uses the English description, problem_en). Download the filtered records as a CSV.
Add / Upload Data	Add your own repair records, one at a time or from a CSV file. See below.
Predict Repair Outcome	Enter a product's details and click Predict to estimate whether it is likely to be Fixed, Repairable or End of life.
Model Evaluation	How accurate the prediction model is. Click Retrain model to include your own records.
Fault Analysis	Groups of similar problem descriptions (fault clusters) and their repair outcomes.
Demand Trends	Monthly repair counts and simple six-month forecasts.

Adding your own data

Open **Add / Upload Data**. It has three tabs:

Add one record: fill in the form (category, brand, status, date and an English problem description) and click **Add record**. A green message confirms it was saved.

Upload CSV file: click **Download CSV template**, open it in Excel, add one row per repair, and save it as **CSV**. Then click **Browse files**, choose your file, check the preview, and click **Add ... record(s)**.

Column	Required?	Example
event_date	Yes	2026-09-14 (year-month-day)
repair_status	Yes	Fixed, Repairable, End of life or Unknown
product_category	Yes (or product_category_id)	Laptop
problem_en	Recommended	Battery not charging (English description)
problem	Optional	Akku lädt nicht (original language)
brand, country, data_provider, year_of_manufacture, product_age, repair_barrier	Optional	Dell, GBR, ..., 2018, 6, ...

Manage added records: tick records and click **Remove selected**, or click **Remove all added records** to go back to the original data.

After adding records, every page updates automatically. Use the **Data source** filter to see only your records (*User added*). To include your records in predictions, open **Model Evaluation** and click **Retrain model**; **Restore bundled model** switches back.

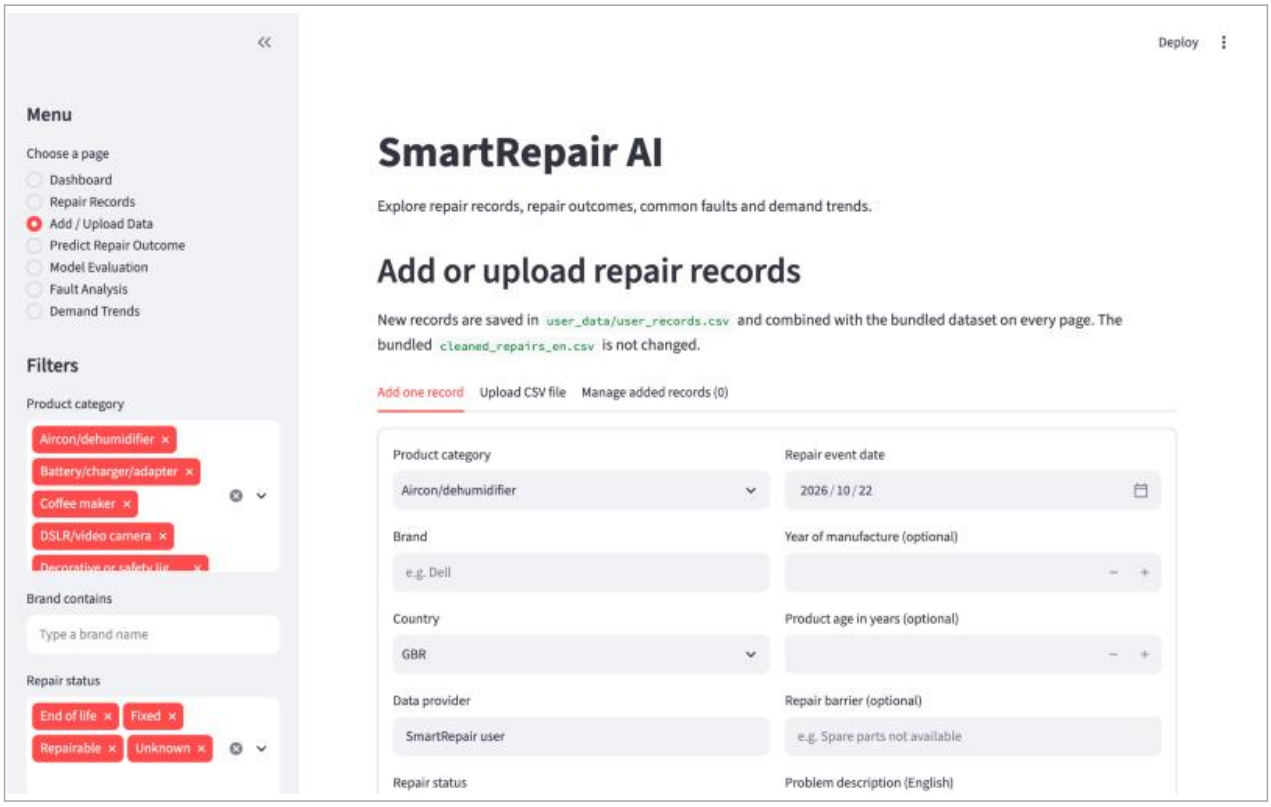


Figure 2: The Add / Upload Data page.

> **Good to know:** Your records are saved in **user_data/user_records.csv**, so they are still there next time you start the app. The original dataset file is never changed. To start completely fresh, stop the app and delete the user_data folder.

7. Stop the app, and run it again next time

To stop: click on the terminal window and press **Ctrl + C** (on macOS, Control + C). You can then close the terminal.

Next time you only need two steps, because Python and the packages are already installed:

1. Open a terminal in the SmartRepair folder (section 3).

2. Start the app (section 5):

Windows (use python3 on macOS)

```
python -m streamlit run app.py
```

8. Troubleshooting

Problem	What to do
'python' is not recognized (Windows)	Python is not on PATH. Re-run the Python installer, choose Modify, tick *Add Python to environment variables*. Close and reopen PowerShell. You can also try `py` instead of `python`.
command not found: python (macOS)	Use `python3` instead of `python`.
'streamlit' is not recognized	Always start the app with `python -m streamlit run app.py` (not just `streamlit run`).
No module named streamlit / pandas / sklearn	The packages are not installed. Repeat section 4, then start the app again.
ERROR: Could not open requirements file	The terminal is in the wrong folder. Repeat section 3 and check that `dir` / `ls` lists requirements.txt.
pip install fails for scikit-learn	Your Python version may be too old or too new. Install Python 3.12 from python.org and try again.
cleaned_repairs_en.csv is missing	A file was renamed, moved or not unzipped. Keep all files from section 1 together in one folder beside app.py.
Port 8501 is already in use	The app is already running in another terminal; use that one, or start a second copy on another port: `python -m streamlit run app.py --server.port 8502`
The browser shows 'This site can't be reached'	The app is not running. Check the terminal for an error, then start it again.
The page is slow or keeps showing 'Running...'	Normal on first load and when you change filters (300,000 records). The first visit to Fault Analysis after adding records can take a few seconds longer.
The prediction model could not be loaded	scikit-learn must be version 1.8.0. Run the install command in section 4 again.
My uploaded CSV shows rows were skipped	Each row needs a valid event_date (YYYY-MM-DD), a repair_status and a product_category. Rows whose id already exists are also skipped. Fix the rows and upload again.

9. Quick reference card

Task	Windows (PowerShell)	macOS (Terminal)
Check Python	python --version	python3 --version
Install packages (once)	python -m pip install -r requirements.txt	python3 -m pip install -r requirements.txt
Check files (optional)	python smoke_test.py	python3 smoke_test.py
Start the app	python -m streamlit run app.py	python3 -m streamlit run app.py
Open in browser	http://localhost:8501	http://localhost:8501
Stop the app	Ctrl + C	Control + C

Predictions and forecasts are estimates based on community repair-event data. They are not guarantees and do not replace a technician's judgement.